

L1: Trivial Networks Versus Complex Networks

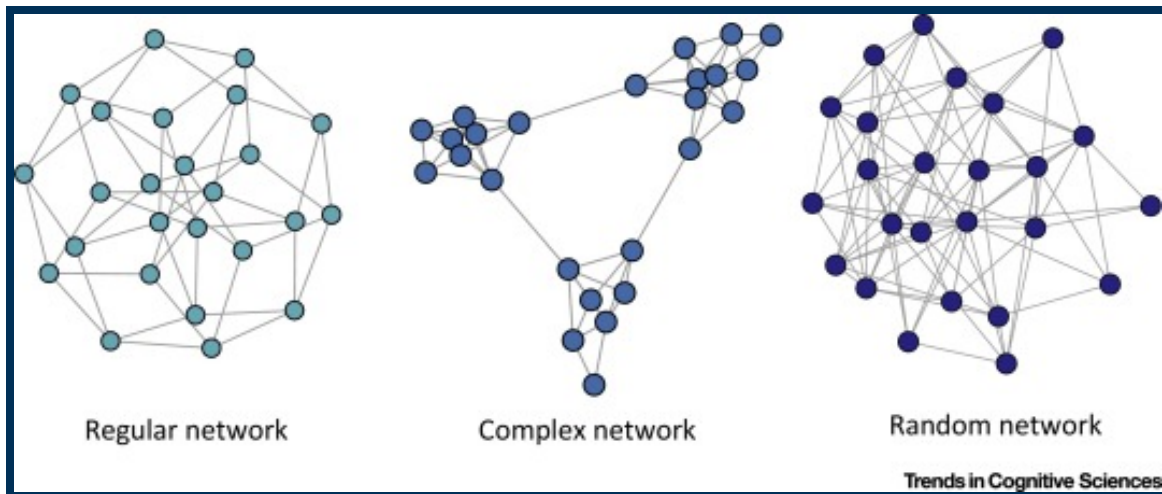


Image Source: [Local Patterns to Global Architectures: Influences of Network Topology on Human Learning. Karuza, Thompson-Schill, Bassett; 2016](https://www.cell.com/trends/cognitive-sciences/fulltext/S1364-6613(16)30071-7) ➞
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Trivial Networks also known as regular or random networks differ significantly from complex networks as the image above shows.

“Regular networks” are a large family of networks that have been studied extensively by mathematicians over the last couple of centuries. Regular networks such as rings, cliques, lattices, etc, have the same interconnection pattern, the same structure, at all nodes.

The example shown at the left in the image above is a regular network in which every node connects to four other nodes.

Another well-studied class of networks in graph theory is that of **“Random networks”**. Here, the connections between nodes are determined randomly. In the simplest model, each pair of nodes is connected with the same probability.

In practice, most technological, biological and information systems do **NOT** have a regular or random network architecture. Instead, their architecture is highly specific, resulting in interconnection patterns that are highly variable across nodes.

For example, the network in the middle has several interesting properties that would not be expected if the network was randomly “*wired*”: note that there are three major clusters of nodes, few nodes have a much larger number of connections than others, and there are many connected three-node groups.

A major difference between network science and graph theory is that the former is an applied data-science discipline that focuses on complex networks encountered in real-world systems.

Graph theory, on the other hand, is a mathematical field that focuses mostly on regular and random graphs. We will return to the connection between these two disciplines later in this lesson.